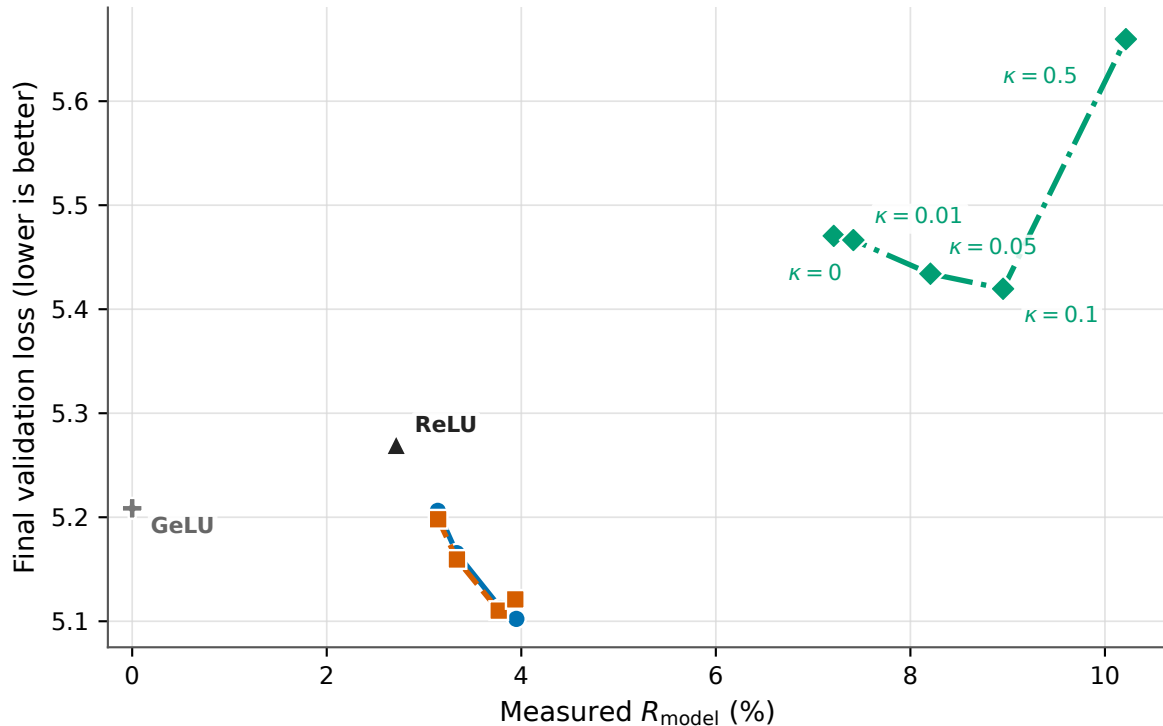


Pythia-14M: validation quality versus logical sparsity opportunity

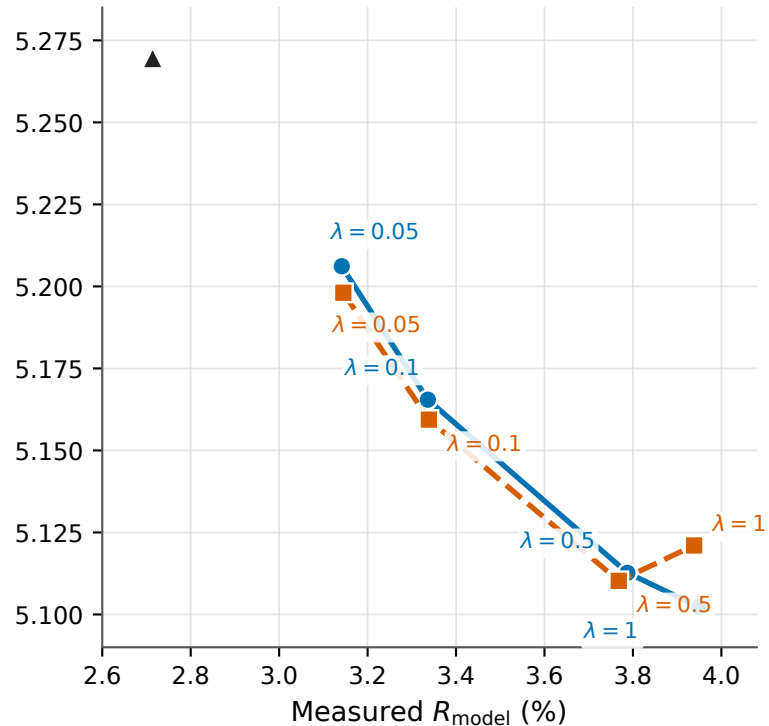
One MiniPile pass, one seed, and all 338 complete validation blocks

—●— A1-H naive L1 —■— A1-H OL1 —◆— A4-Z threshold + GeLU control ▲ A1-H ReLU control

(a) Full quality-opportunity range



(b) A1-H endpoint detail



Lines connect increasing dose only. Validation-loss axes show endpoint detail and do not start at zero.
 R_{model} is exact-zero logical-product opportunity, not measured speedup; no uncertainty bars (one seed).